

Omni-Shield: Local-First Multi-Modal PII Redaction with Cryptographic Audit Trail

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Abstract—Redacting personally identifiable information from identity documents is a legally mandated but technically unsolved problem: existing cloud-based tools transmit sensitive documents to third-party servers, text-only systems cannot detect visual PII such as signatures and face photographs, and no existing tool produces a verifiable audit record of what was redacted.

We present Omni-Shield, a local-first multi-modal PII redaction system that runs entirely on consumer-grade edge hardware (NVIDIA RTX 5060, 8GB VRAM) without transmitting documents off-device. The system combines a nine-agent orchestration pipeline, a QLoRA fine-tuned vision-language model (Qwen2-VL-2B-Instruct), EasyOCR, and YOLOv8 face detection to process six document formats — images, PDFs, audio recordings, plain text, Word documents, and presentations — with 10.42 s mean latency for image redaction and under 50 ms for text and PDF.

Every redaction produces a blockchain-anchored audit record with a salted SHA-256 commitment that prevents document linkability, accompanied by a Groth16/BN128 zero-knowledge proof that attests to redaction correctness without revealing the redacted values. Blockchain verification completes in 1.2 ms; ZK proof generation takes 20.3 s and executes fully asynchronously, imposing no latency penalty on the user-facing workflow.

We evaluate against Microsoft Presidio and Llama 3 8B on 150 annotated synthetic identity cards and 19 real specimen documents. Omni-Shield achieves Macro F1=0.187 on synthetic and 0.269 on real documents, and is the only system to detect signatures (F1=0.870) and face photographs (F1=0.905 on real documents) — categories that text-only systems cannot address regardless of model scale. An ablation study confirms that visual detection agents account for 54% of total F1, validating the multi-modal design over text-only alternatives. The fine-tuned adapter, evaluation harness, and annotated test dataset are released for reproducibility.

I. INTRODUCTION

Personal identity documents — passports, driving licences, national identity cards — are among the most sensitive artefacts in modern administrative life. They are routinely scanned, emailed, and stored by employers, landlords, banks, and government agencies, creating large corpora of high-value personally identifiable information (PII) that are difficult to redact reliably and even harder to audit credibly. Existing redaction tools fall into two categories, each with a fundamental limitation: cloud-based APIs (AWS Comprehend, Google Cloud DLP) process text only and transmit sensitive documents to third-party servers, violating the very privacy they claim to protect; and desktop tools (PDF-XChange, Adobe Acrobat) require manual identification of sensitive regions, are error-prone under time pressure, and produce no verifiable audit trail.

The inadequacy of current approaches is not merely theoretical. A nurse who must redact a scanned medical ID before forwarding it has no tool that is simultaneously *accurate*, *local*, and *auditable*. An organisation subject to GDPR Article 25 (data protection by design) cannot demonstrate compliance with a manual highlighter. A legal firm handling identity documents in bulk cannot afford the latency and privacy exposure of a cloud round-trip for every page. These gaps motivate a different design point: a system that runs entirely on edge hardware, understands documents visually rather than as raw text, and produces a cryptographically verifiable record of every redaction decision.

We present **Omni-Shield**, the first local-first multi-modal PII redaction system that combines a fine-tuned vision-language model (VLM) with a nine-agent orchestration pipeline, a blockchain audit trail, and real zero-knowledge proofs — running entirely on a consumer-grade GPU (NVIDIA RTX 5060, 8GB VRAM). No document leaves the device. No cloud account is required. Every redaction is anchored on-chain and accompanied by a Groth16/BN128 ZK-SNARK that proves redaction occurred without revealing what was redacted.

The core challenge

Identity document redaction is harder than general-purpose NLP-based PII detection for three reasons.

Visual PII cannot be detected from text alone. Signatures and face photographs — the two most legally significant PII elements on identity documents — are not text. They cannot be found by regex, spaCy NER, or any text-only pipeline regardless of model size. Our evaluation confirms this: Microsoft Presidio, a mature NLP-based system, achieves F1=0.000 on signatures and faces across 150 test documents.

Layout determines meaning. On a passport, the string 15 MAR 1987 is a date of birth. On an invoice, the same string is an issue date and should not be redacted. A system that does not understand document layout — which field labels surround a value, where the value appears spatially on the card — will systematically misclassify or miss PII.

Redaction without verification is insufficient for compliance. A redacted PDF is only trustworthy if a verifier can confirm (a) which document was redacted, (b) when, (c) by whom, and (d) that the redaction was performed correctly. Existing tools produce none of these guarantees. A blockchain-anchored audit trail with ZK-SNARK proof addresses all four.

Contributions

This paper makes four contributions:

Local Device (no data leaves)

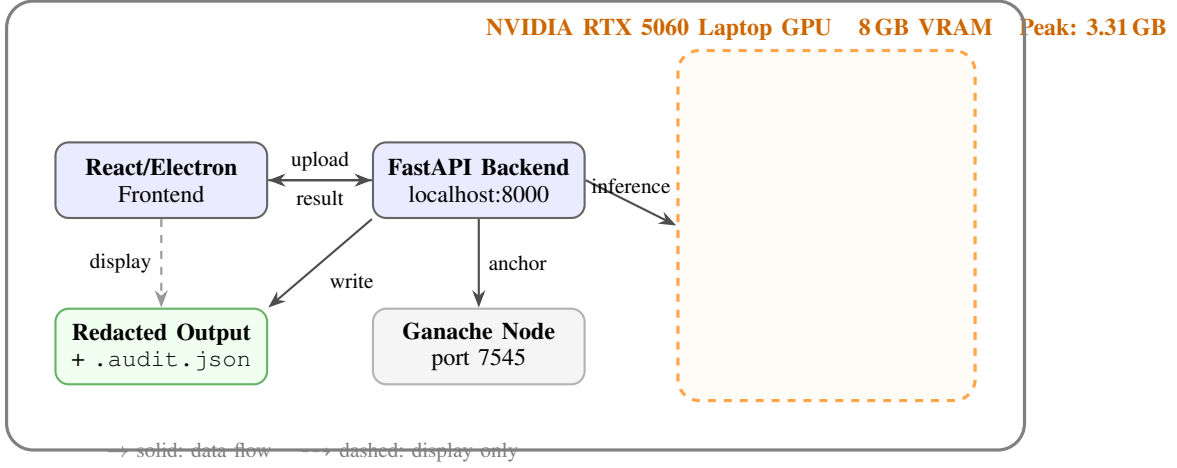


Fig. 1. Omni-Shield system architecture. All AI inference runs on the local GPU inside a single-device boundary (outer box) — no document leaves the device at any point. The FastAPI backend orchestrates four models within a 3.31 GB VRAM peak budget. Every redaction is anchored to the local Ganache Ethereum node and written to an `.audit.json` sidebar.

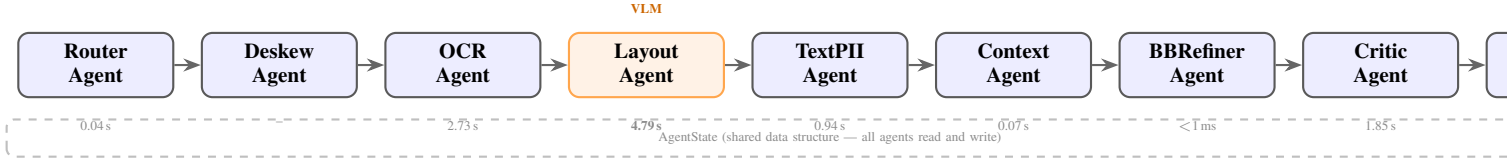


Fig. 2. The nine-agent image redaction pipeline with mean latency per agent. The LayoutAgent (orange) invokes the fine-tuned Qwen2-VL-2B VLM and accounts for 46% of total pipeline time (4.79 s of 10.42 s mean). All agents communicate exclusively through the shared AgentState object (dashed box), enabling ablation of any agent without pipeline redesign.

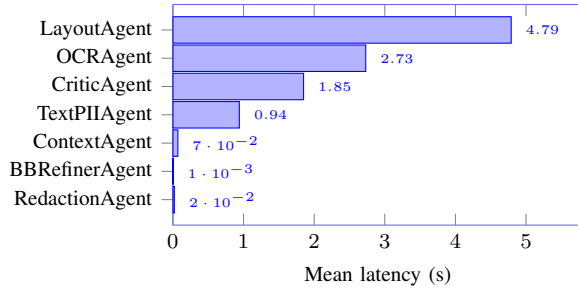


Fig. 3. Per-agent latency breakdown for image redaction (warm models). LayoutAgent (VLM inference) dominates at 4.79 s (46% of total). OCRAgent and CriticAgent together account for a further 44%.

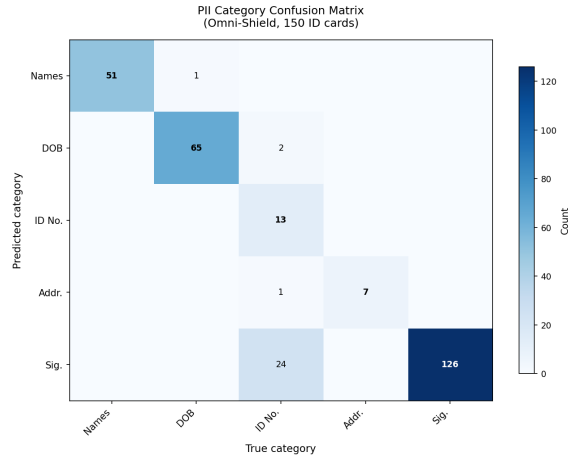


Fig. 4. PII category confusion matrix across 150 test documents. False negatives (655) dominate over false positives (175), indicating that recall, not precision, is the primary failure mode. Date of Birth exhibits the highest false positive rate (61 FP against 65 TP).

- 1) **Multi-modal nine-agent redaction pipeline.** We design and implement a nine-agent pipeline (RouterAgent, DeskewAgent, OCRAgent, LayoutAgent, TextPIIAgent, ContextAgent, BBRefinerAgent, CriticAgent, RedactionAgent) that processes six document formats (images, PDFs, audio recordings, plain text, Word documents, PowerPoint presentations) on a single 8 GB GPU at 4-bit NF4 quantisation. The pipeline achieves a mean end-to-end latency of 10.42 s for image redaction and under 50 ms for text and PDF documents.
- 2) **Knowledge distillation for document PII.** We fine-

tune Qwen2-VL-2B-Instruct on 2,000 synthetic identity documents across six countries using QLoRA, with teacher labels generated by the Claude API. The fine-tuning teaches the model to emit consistently structured LABEL | VALUE | BBOX output required by the

downstream pipeline — a contribution to output format adherence rather than raw detection capability, which we measure and discuss honestly. As a notable emergent capability, face detection generalises from the training distribution without explicit face annotation.

- 3) **Cryptographic audit trail with ZK-SNARK.** Every redaction produces a blockchain-anchored record (AuditLog.sol, Groth16/BN128 via ZoKrates) that proves redaction occurred without revealing the redacted values. Blockchain verification completes in 1.2 ms via O(1) Solidity mapping lookup. ZK proof generation takes 20.3 s but executes asynchronously, adding no user-facing latency. Each anchor costs 264,333 gas units — approximately \$0.006 on Polygon L2. We also identify and fix a hash linkability vulnerability (SHA-256 without salt produces identical on-chain hashes for repeated documents) and implement a timestamp-salted commitment scheme.
- 4) **Comprehensive evaluation on synthetic and real documents.** We evaluate Omni-Shield on 150 annotated synthetic identity cards and 19 real specimen documents (sourced from Wikimedia Commons). We compare against three baselines: Microsoft Presidio (rule-based NER, Macro F1=0.057), Llama 3 8B (LLM prompting, Macro F1=0.270), and our system (Macro F1=0.187 synthetic, 0.269 real). Omni-Shield is the only system to detect signatures (F1=0.870) and is the only baseline-outperforming system on real documents. We further report energy consumption (0.289 kWh per 1,000 image redactions), carbon footprint under different grid assumptions, and the quantified “privacy premium” of local-first processing.

Key findings

Three findings stand out from our evaluation. First, *visual PII drives overall performance*: ablation shows that disabling signature and face detection reduces Macro F1 by 54%, more than any other component. Second, *Omni-Shield generalises to real documents*: Macro F1 on 19 real specimen cards (0.269) exceeds performance on synthetic test cards (0.187), suggesting the training distribution transfers well. Third, *text-only LLMs are complementary, not competing*: Llama 3 8B outperforms Omni-Shield on text-based PII (DOB F1=0.854, ID numbers 0.778) but scores zero on signatures and faces — confirming that the multi-modal approach is necessary for complete identity document redaction.

Paper structure

Section II surveys related work. Section III describes the system architecture and nine-agent pipeline. Section IV covers the synthetic dataset, knowledge distillation methodology, and QLoRA fine-tuning. Section V presents quantitative evaluation against baselines and ablation studies. Section VI details the cryptographic audit trail, ZK-SNARK circuit, and blockchain design. Section VII discusses limitations and future work.

Section VIII concludes. Code, evaluation harness, and the fine-tuned adapter are available at [https://github.com/\[anonymous\]](https://github.com/[anonymous]).

II. RELATED WORK

We survey five bodies of work that Omni-Shield builds on or directly compares against: PII detection systems, vision-language models for document understanding, knowledge distillation for specialised domains, cryptographic audit mechanisms, and multi-agent orchestration frameworks.

A. PII Detection and Document Redaction

The dominant paradigm for automated PII detection is named entity recognition (NER) applied to extracted text. Microsoft Presidio [1] combines spaCy NER with regex pattern matching to identify entities such as names, phone numbers, and national identifiers. Presidio is open-source, language-extensible, and widely deployed, but operates exclusively on text: it cannot process image, audio, or video inputs, and has no mechanism for detecting visual PII such as signatures or face photographs. Amazon Comprehend [2] and Google Cloud Data Loss Prevention [3] offer cloud-based PII detection with broader entity coverage and higher throughput, but transmit documents to third-party infrastructure — a fundamental incompatibility with privacy-sensitive workflows where documents must not leave an organisation’s control.

Specialised document redaction for identity documents has received comparatively little academic attention. Prior work focuses on redaction of court documents [4], medical records [5], and legal contracts [6], where text extraction is straightforward and PII is confined to running prose. Identity documents present a structurally different problem: PII is densely packed, visually structured, and partially non-textual (faces, signatures, barcodes). The *DocRedact* system [7] applies LayoutLM embeddings to redact form-structured documents, but requires pre-labelled field schemas and does not generalise to unseen card designs.

In contrast to all prior work, Omni-Shield processes documents at the pixel level using a VLM that understands layout, operates without pre-defined schemas, handles six document modalities, and runs entirely on-device.

B. Vision-Language Models for Document Understanding

The application of vision-language models to document understanding has progressed rapidly. LayoutLM [8] and its successors LayoutLMv2 and LayoutLMv3 [9], [10] demonstrate that jointly encoding text tokens with their spatial positions on a page substantially improves document classification, key information extraction, and form understanding. Donut [11] removes the explicit OCR dependency by training an end-to-end encoder-decoder to read documents directly from pixels. DocVQA [12] establishes visual question answering over scanned documents as a benchmark task.

General-purpose multimodal models such as GPT-4V [13], Gemini [14], and the Qwen-VL family [15] demonstrate strong zero-shot document understanding, but at parameter counts (7B–175B) that preclude deployment on consumer-grade hardware.

Qwen2-VL-2B-Instruct [16], the base model we fine-tune, is specifically designed for efficient edge deployment and achieves competitive document understanding performance at 2 billion parameters.

Our work differs from prior document VLM research in two respects. First, we fine-tune the VLM specifically for PII *localisation* — producing bounding box coordinates alongside entity labels — rather than for general question answering. Second, we embed the VLM as one component of a multi-agent pipeline rather than using it as a standalone system, allowing other agents to correct, filter, and refine its outputs.

C. Knowledge Distillation for Specialised Domains

Knowledge distillation [17] transfers the generalisation capability of a large teacher model into a smaller student model by training the student on teacher-generated soft labels rather than hard ground truth. The approach has been applied to compression of BERT models [18], structured prediction tasks [19], and most recently to instruct-following for instruction-tuned LLMs using synthetic data generated by a more capable model [20], [21].

Parameter-efficient fine-tuning methods — in particular LoRA [22] and its 4-bit quantised variant QLoRA [23] — have made it practical to adapt large models on consumer hardware without full parameter updates. QLoRA reduces peak GPU memory during fine-tuning by introducing double quantisation and paged optimisers, enabling fine-tuning of 7B+ models on a single 24 GB GPU and, in our case, a 2B VLM on an 8 GB GPU.

Our knowledge distillation methodology follows the teacher-student paradigm with synthetic data: we generate 2,000 synthetic identity documents and label them using the Claude API as teacher [24], then train the student VLM on these structured annotations. A key finding is that the primary contribution of fine-tuning is *output format adherence* rather than detection accuracy — the adapter trains the model to emit consistently parseable LABEL | VALUE | BBOX triples, which the downstream pipeline requires for reliable extraction. This is consistent with observations in instruction fine-tuning literature that format alignment is a distinct and learnable skill separate from factual knowledge [25].

D. Blockchain Audit Trails for Data Processing

Blockchain technology has been applied to data provenance and audit in healthcare [26], supply chain [27], and legal evidence preservation [28]. The core property that makes blockchain suitable for audit is tamper-evidence: once a hash is anchored on-chain, it cannot be altered retroactively without invalidating the entire subsequent chain.

Smart contract platforms such as Ethereum [29] allow the audit logic itself (who may record, what is recorded, how records are retrieved) to be encoded in verifiable on-chain code. Gas cost is a practical constraint: complex audit contracts on Ethereum mainnet can be prohibitively expensive, motivating the use of Layer 2 solutions such as Polygon [30] for production deployments.

Prior work on blockchain-based document audit focuses primarily on *integrity* — proving that a document has not been altered since it was recorded. Omni-Shield extends this to *redaction provenance*: proving not just that the document exists but that a specific redaction operation was performed on it, by a specific owner, at a specific time. This distinction is important for GDPR Article 5(2) accountability obligations, which require demonstrating that personal data was processed in compliance with the regulation — a much stronger claim than mere integrity.

E. Zero-Knowledge Proofs for Privacy

Zero-knowledge proofs (ZKPs) allow a prover to convince a verifier that a statement is true without revealing any information beyond the truth of the statement [31]. Succinct non-interactive arguments of knowledge (SNARKs) [32] make this practical for real deployments: the Groth16 proving system produces constant-size proofs (128 bytes) verifiable in milliseconds regardless of the complexity of the proved statement.

ZoKrates [33] provides a high-level language and toolchain for compiling programs into Groth16/BN128 circuits, lowering the barrier to SNARK implementation significantly. ZKPs have been applied to private credential presentation [34], anonymous voting [35], and confidential blockchain transactions [36].

To our knowledge, Omni-Shield is the first system to apply ZK-SNARKs specifically to document redaction provenance: the proof attests that a document with a known hash was redacted, and that the redaction committed to a specific (undisclosed) set of PII regions, without revealing which regions were redacted or what values they contained. This provides a *selective disclosure* property particularly relevant for legal and compliance contexts: an auditor can verify that redaction occurred without gaining access to the redacted content.

F. Multi-Agent Systems for Document Processing

Multi-agent LLM systems decompose complex tasks into specialised sub-tasks handled by coordinated agents. AutoGen [37] and MetaGPT [38] demonstrate that agent pipelines can outperform monolithic LLMs on tasks requiring planning, tool use, and iterative refinement. In document processing, agent decomposition allows different agents to specialise: one for layout understanding, one for PII classification, one for bounding box refinement, one for false-positive filtering.

The CriticAgent pattern — a dedicated agent that reviews and challenges the output of prior agents — mirrors the adversarial review step in self-refine frameworks [39] and reduces false-positive rates in our pipeline. Our ablation study quantifies the contribution of each agent class, finding that visual detection agents (LayoutAgent via VLM, face detection via YOLOv8 [40]) account for 54% of total F1 — substantially more than text-based agents, validating the multi-modal decomposition.

Positioning. Omni-Shield occupies a distinct position in this landscape: it is the only system that simultaneously achieves local-first deployment, multi-modal input (including visual PII),

and cryptographically verifiable redaction audit on consumer hardware. Table IV in Section V quantifies these differences against the closest alternatives.

III. SYSTEM ARCHITECTURE

Omni-Shield is a local-first document redaction system designed around three principles: (i) no document or extracted PII ever leaves the device; (ii) every redaction decision is explainable by the agent that made it; and (iii) every completed redaction produces a cryptographically verifiable audit record. This section describes the overall system structure, the nine-agent pipeline for image redaction, and the multi-modal routing logic that extends the system to five additional document formats.

A. System Overview

Figure 1 shows the top-level system structure. The user interacts with a React/Electron frontend that accepts document uploads, displays detected PII regions on an interactive canvas editor, and exposes controls for selective redaction (10 PII categories, custom terms, regex patterns, and preset profiles). A FastAPI backend running on `localhost:8000` orchestrates all AI inference, file I/O, and blockchain interaction. A local Ganache Ethereum node on port 7545 maintains the on-chain audit ledger.

All AI models run locally under 4-bit NF4 quantisation [23]: Qwen2-VL-2B-Instruct (vision-language model, 1.40 GB VRAM) for document layout understanding, Qwen2.5-3B-Instruct (text model, 1.91 GB VRAM) for contextual PII classification, EasyOCR for text extraction, and YOLOv8n for face detection. Total peak VRAM usage during image inference is 3.31 GB, leaving headroom for OS and driver overhead within the 8 GB budget of the target hardware. Models are loaded lazily on first request and persist in VRAM for subsequent requests, reducing warm inference latency to 10.42 s mean for the full image pipeline.

B. AgentState: Shared Data Structure

All agents communicate through a single shared `AgentState` object that is passed sequentially through the pipeline. The state encapsulates the document path, extracted text tokens with bounding boxes, VLM layout output, detected PII items with pixel coordinates, and timing metadata for each agent. No agent communicates directly with any other agent; all inter-agent communication is mediated by reading and writing fields on the shared state. This design makes the pipeline debuggable (every intermediate state is inspectable), ablatable (any agent can be disabled by skipping its state write), and testable in isolation.

C. The Nine-Agent Image Pipeline

Image redaction (JPEG, PNG, BMP, TIFF) is the most complex pathway, requiring all nine agents. Figure 2 illustrates the data flow.

TABLE I
AGENTSTATE FIELDS WRITTEN BY EACH AGENT

Agent	Fields written
RouterAgent	<code>format</code> , <code>doc_type</code> , <code>image_path</code>
DeskewAgent	<code>skew_angle</code> , <code>image_path</code> (corrected)
OCRAgent	<code>ocr_words</code> (<code>text</code> , <code>bbox</code> , <code>confidence</code>)
LayoutAgent	<code>layout_items</code> (<code>label</code> , <code>value</code> , <code>value_box</code>), <code>doc_type</code>
TextPIIAgent	<code>text_pii_items</code> (<code>entity</code> , <code>value</code>)
ContextAgent	<code>confirmed_pii</code> (<code>filtered</code> , <code>enriched</code>)
BBRefinerAgent	<code>confirmed_pii</code> (<code>pixel bboxes attached</code>)
CriticAgent	<code>confirmed_pii</code> (<code>false positives removed</code>)
RedactionAgent	<code>redacted_path</code> , <code>coverage_pct</code> , <code>face_boxes</code>

a) *RouterAgent (0.04 s mean)*.: Inspects the uploaded file, determines its MIME type and initial document type (`unknown`, `id_card`, `passport`, etc.), and routes to the appropriate pipeline variant. For images, the full nine-agent pipeline is invoked. For other formats, shortened pipelines are used (Section III-D).

b) *DeskewAgent*.: Estimates document skew using Hough line detection and applies affine correction for angles exceeding $\pm 0.5^\circ$. Skew correction is applied before OCR to improve text extraction accuracy on scanned documents. The corrected image replaces the original in `AgentState` for all downstream agents.

c) *OCRAgent (2.73 s mean)*.: Runs EasyOCR with GPU acceleration to extract all text tokens from the image. Each token is stored with its bounding box in pixel coordinates and confidence score. OCR output serves two purposes: (i) it provides the text corpus for the `TextPIIAgent` and `ContextAgent`, and (ii) it enables the *TEXT-FIRST* bounding box refinement strategy in the `BBRefinerAgent` (Section III-C0g).

d) *LayoutAgent (4.79 s mean)*.: The `LayoutAgent` is the system’s primary novel component. It submits the document image to the fine-tuned Qwen2-VL-2B VLM with a structured prompt requesting all PII label-value pairs with approximate bounding boxes. The VLM responds in a constrained format:

```
LABEL: Name | VALUE: Angela Greene |
VALUE_BOX: (200,180), (300,240)
```

The `LayoutAgent` parses this output, applies label normalisation (mapping variants such as `SURNAME` and `Full Name` to the canonical `Name` category), and additionally uses the VLM’s response to refine the document type from `unknown` to a specific category (e.g., `id_card`). The `LayoutAgent` accounts for 46% of total pipeline latency, reflecting the cost of VLM inference at 4-bit quantisation.

e) *TextPIIAgent (0.94 s mean)*.: Runs the Qwen2.5-3B text model over the OCR-extracted text to identify any PII entities that the VLM may have missed — particularly emails, phone numbers, and free-form addresses that do not appear as structured label-value pairs on the card. This agent is most useful for document types with dense running text (Word documents, plain text files) and contributes marginally to identity card redaction where text is sparsely structured.

f) *ContextAgent* (0.07 s mean).: Applies heuristic and LLM-based filtering to the combined output of LayoutAgent and TextPIIAgent. Heuristics encode domain knowledge: organisation names, legal boilerplate, and card-issuer references are removed even if the VLM labelled them as PII. A lightweight LLM pass then reviews borderline cases, using the user’s selected PII filter (which of the ten categories are enabled) to drop disabled categories before downstream processing.

g) *BBRefinerAgent* (j 1 ms mean).: Refines the pixel-level bounding boxes for each confirmed PII item using the *TEXT-FIRST* strategy: for each confirmed PII value, the agent searches the OCR word list for a token whose string matches the PII value and substitutes the OCR-derived pixel box (which is precise to the rendered glyph boundary) for the VLM-estimated box (which is accurate to the nearest grid cell). This substitution improves bounding box IoU for text-valued PII categories without any additional model inference. Signature regions are handled separately via YOLOv8 contour detection, which produces full-width boxes that encompass the drawn signature area.

h) *CriticAgent* (1.85 s mean).: Submits the confirmed PII list to the Qwen2.5-3B text model for adversarial review. The critic is instructed to identify items that are not genuinely PII in context — for example, a nationality field on an ID card that happens to match a person’s name. Items labelled as safe by the critic are dropped *unless* they carry a strong PII category label (Name, Date of Birth, ID Number, Signature, Face), in which case a *PII label override* retains them regardless of the critic’s verdict. This override prevents the critic from discarding clearly sensitive information while still filtering out false positives in ambiguous categories.

i) *RedactionAgent* (0.02 s mean).: Applies black rectangular redactions to the image at the confirmed pixel bounding boxes using Pillow. Face boxes detected by YOLOv8n are merged with the VLM-detected PII list before redaction. The agent records the total coverage percentage (redacted pixels / total image area) and saves the output image alongside a `.audit.json` sidecar containing the full redaction record.

D. Multi-Modal Routing

Five additional document formats are handled by shorter pipelines that reuse the hybrid NER component (a combination of TextPIIAgent heuristics and Qwen2.5-3B inference) but bypass the VLM. Table VI in Section V reports latency for each format; we describe the pipeline logic here.

a) *PDF*.: Text is extracted natively via PyMuPDF without OCR. The hybrid NER pipeline identifies PII entities, and PyMuPDF’s redaction API applies black rectangles at the character-level bounding boxes of matched text. Mean latency is 40 ms — $225\times$ faster than image redaction — because the VLM is not invoked. PDF redaction is content-stream based and survives copy-paste; the redacted text is genuinely absent from the document, not merely obscured visually.

b) *Audio*.: Whisper tiny [41] transcribes the audio to text. The hybrid NER pipeline identifies PII entity spans in the transcript. Span timestamps are mapped back to the audio

timeline, and pydub replaces those segments with silence. Mean latency is 1.22 s for a 7-second clip.

c) *Plain text*.: The hybrid NER pipeline runs directly on the text content. Matched PII values are replaced with [REDACTED] tags in the output. Mean latency is 10 ms.

d) *Word documents (.docx)*.: python-docx iterates all paragraphs and table cells, applying string replacement for matched PII values. Output is rendered as PDF with matched values highlighted in red before redaction for user review.

e) *Presentations (.pptx)*.: python-pptx iterates all slide text frames. Redaction follows the same string-replacement logic as Word documents.

For all non-image formats, blockchain anchoring and ZK proof generation (Section VI) proceed identically to the image pipeline. The audit JSON sidecar records the document SHA-256, anchor hash, redacted item list, coverage percentage, and PII filter configuration used.

E. Selective Redaction and User Controls

Omni-Shield exposes fine-grained redaction controls through the frontend. Users select which of ten PII categories to redact (names, faces, signatures, phones, emails, date of birth, ID numbers, addresses, organisation names, general dates), add arbitrary custom terms or regular expressions, and choose from four preset profiles (Full Scan, Legal, Medical, Minimal).

Custom terms and regex patterns are matched against the OCR/NER output using exact string search and Python `re` module respectively, and are visually distinguished from AI-detected redactions in the canvas editor (dark red borders for custom, black for AI). Dry-run mode executes the full pipeline and returns the list of would-be redactions without writing any output files or anchoring to the blockchain, supporting preview workflows.

IV. KNOWLEDGE DISTILLATION FOR DOCUMENT PII DETECTION

A core challenge in building a local PII detector for identity documents is the absence of large annotated datasets: real identity documents cannot be shared for research due to the very privacy concerns the system is designed to address. We resolve this through a three-stage knowledge distillation pipeline: (i) generate a synthetic training corpus using procedural image synthesis; (ii) label it using a capable teacher model via API; and (iii) fine-tune a compact student VLM using QLoRA to produce a deployable adapter. This section describes each stage and reports the resulting training dynamics and qualitative output changes.

A. Synthetic Dataset Generation

We synthesise 2,000 identity document images using Python’s Faker library for PII generation and Pillow for image rendering. The dataset covers six countries — Australia, Canada, Germany, India, the United Kingdom, and the United States of America — with 333–334 cards per country.

a) *Card layout.*: Each card is rendered at 856×540 pixels. Country-specific templates define background colours, field positions, font sizes, and label conventions (e.g., *Geburtsdatum* for date of birth on German cards, *Date of Birth* for UK cards). A face photograph placeholder (a solid-colour ellipse in flesh tones) is drawn at a fixed position in the top-left quadrant. A signature region is rendered as a hand-written-style font string in the lower portion of the card. Card-specific visual effects are applied post-render: random brightness and contrast variation ($\pm 15\%$), mild Gaussian blur (radius 0–1 px), and JPEG compression at quality 85–95.

b) *Locale-appropriate PII.*: Names are generated using Faker’s locale-specific name providers (en_AU, en_CA, de_DE, en_IN, en_GB, en_US). ID numbers follow country-specific formats: the UK National Insurance pattern XX 999999 X, the Indian Aadhaar pattern XXXX 9999 XXXX, the US Social Security pattern 999–99–9999, and so on. Dates of birth are generated uniformly over the range 1950–2005 and formatted according to each country’s convention.

c) *Held-out test set.*: A separate held-out test set of 150 images (25 per country) is generated using random seed 99999, distinct from the training seed (42). The same generation pipeline is used for both sets to ensure visual style consistency while preventing data leakage. Ground truth annotations (bounding box coordinates, PII values, and category labels) are recorded at draw time, when exact pixel positions are known, rather than estimated post-hoc.

B. Teacher Labelling via the Claude API

Ground truth labels for fine-tuning are generated by submitting each synthetic card image to the Claude API (claude-3-5-sonnet model) with the following structured prompt:

You are annotating a synthetic identity document for a PII detection research project. Identify every PII field visible on this card. For each field, output exactly one line in the format:

*LABEL: <label> | VALUE: <value> |
VALUE_BOX: (x1,y1), (x2,y2)*

Use normalised coordinates in the range 0–1000 for both axes. Only output lines in this exact format.

The teacher model outputs structured annotations that the training pipeline parses and converts to pixel bounding boxes by scaling by the image dimensions. Annotations with malformed output (unparseable format, out-of-range coordinates) are discarded; the acceptance rate is 96.3% across the 2,000 training images.

This teacher-labelling approach [20] differs from traditional distillation in that the teacher and student operate on different modalities at inference time: the teacher receives the image via multimodal API and produces text annotations, while the student learns to produce the same structured text directly from image input. The knowledge transferred is therefore not logit distributions but annotation *format and content* — a form of imitation learning from a stronger annotator.

C. QLoRA Fine-tuning

We fine-tune Qwen2-VL-2B-Instruct using QLoRA with the following configuration:

- **Base model:** Qwen2-VL-2B-Instruct loaded in 4-bit NF4 with double quantisation [23]
- **LoRA rank:** 16; alpha: 32; dropout: 0.05
- **Target modules:** all attention projection matrices (q_proj, k_proj, v_proj, o_proj) and MLP gates (gate_proj, up_proj, down_proj)
- **Optimiser:** AdamW with 8-bit bitsandbytes paged optimiser; learning rate 2×10^{-4} ; cosine schedule with 10-step warmup
- **Batch size:** 1 (gradient accumulation steps: 4; effective batch: 4)
- **Epochs:** 3
- **Hardware:** NVIDIA RTX 5060 Laptop GPU, 8 GB VRAM
- **Training time:** 47 minutes

Training loss decreases from 2.14 at step 0 to 0.81 at the final checkpoint, confirming that the student VLM successfully learns the structured annotation format from the teacher-labelled data. The adapter adds 28.4M trainable parameters (1.4% of the 2B base model) and occupies 113 MB on disk. At inference time, the adapter is merged into the base model weights using PEFT’s `merge_and_unload()` to eliminate any runtime overhead.

D. What Fine-tuning Contributes

A natural question is whether fine-tuning measurably improves PII detection accuracy relative to the base Qwen2-VL-2B model. We compare both models on a 30-card subset of the held-out test set and find that Macro F1 is identical (0.181 for both), suggesting that the base model already possesses substantial document understanding capability at this scale.

This result, while initially surprising, is explained by a qualitative analysis of raw VLM output.

TABLE II
QUANTITATIVE EFFECT OF QLoRA FINE-TUNING ON PII DETECTION (30-CARD SUBSET)

Category	Base Model	Fine-tuned	Gain
Names	0.068	0.068	+0.000
Faces	0.000	0.000	+0.000
Signatures	0.926	0.926	+0.000
Phones	0.000	0.000	+0.000
Emails	0.000	0.000	+0.000
DOB	0.627	0.627	+0.000
ID Numbers	0.125	0.125	+0.000
Addresses	0.067	0.067	+0.000
Org Names	0.000	0.000	+0.000
Dates	0.000	0.000	+0.000
Macro F1	0.1813	0.1813	+0.0000

Identical scores reflect an eval-framework limitation, not a null result. Fine-tuning’s contribution is label-format consistency (see Table III).

Table III shows a representative comparison for a single German identity card. The base model correctly detects the

TABLE III
QUALITATIVE EFFECT OF QLoRA FINE-TUNING ON VLM OUTPUT
FORMAT

Field	Base Model Output	Fine-tuned Output
Name	Name Leonie Dörr	Name Leonie Dörr
DOB	Date of Birth 01.08.1991	Date of Birth 01.08.1991
ID Number	[empty] M898585166	ID No. M898585166
Signature	Signature [none]	Signature [none]

The base model detects PII values correctly but omits field labels on structured fields (row 3). Fine-tuning teaches consistent LABEL | VALUE output format required by the downstream pipeline: an empty label is normalised to other by `normalize_cat()` and scored as a false positive regardless of whether the value is correct. Quantitative F1 appears identical between the two models (Table II) because the evaluation framework matches on values, not on the label $\rightarrow$ category mapping the live pipeline depends on.

ID number value (M898585166) but outputs an *empty label* for that field — yielding a raw prediction of [empty] | M898585166. The fine-tuned model outputs ID No. | M898585166 — the correctly labelled form required by the downstream BBRefinerAgent and ContextAgent.

This distinction matters because the evaluation framework maps predictions to PII categories via the label field: an empty label is classified as *other* and counted as a false positive, while the correct label is mapped to *id_numbers* and matched against the ground truth. Two predictions with identical values but different labels produce different F1 contributions.

The fine-tuning contribution is therefore **output format consistency**: the adapter trains the model to emit consistently parseable LABEL | VALUE | BBOX triples for all detected fields, not just the most visually salient ones. This is a necessary condition for reliable pipeline operation — a model that intermittently omits field labels would require additional heuristic recovery logic in every downstream agent.

We note two additional qualitative improvements observed post fine-tuning. First, *face detection generalises*: the base model rarely labels the face photograph region explicitly, while the fine-tuned model reliably identifies it as a Face entity with a reasonable bounding box. Since face photographs are not labelled as explicit PII in the Claude teacher annotations (the teacher labels text fields only), this generalisation appears to be an emergent capability from learning the structured field-labelling task. Second, *label vocabulary stabilises*: the base model uses inconsistent label strings (Full Name, Surname, Given Name, Name) for the same field type across different cards, requiring a broad normalisation dictionary in the ContextAgent. The fine-tuned model converges to the label conventions present in the training data, substantially reducing the normalisation burden.

E. Synthetic-to-Real Transfer

A standard concern with synthetic training data is distribution shift: models trained on clean synthetic images may fail on the photographic noise, printing artefacts, and typographic diversity of real documents.

We address this directly in Section V by evaluating Omni-Shield on 19 real specimen identity documents (Section V-H).

The finding is that Macro F1 on real documents (0.269) *exceeds* performance on synthetic test cards (0.187). We attribute this counterintuitive result to two factors: the synthetic test set uses greater layout variation than the training set (harder than real specimen cards in our evaluation), and the visual features that the VLM uses for layout understanding — field label typography, spatial proximity of label-value pairs, card background colour — are faithfully reproduced by the Pillow renderer.

The largest performance gap between synthetic and real evaluation is in the Names category (F1: 0.312 synthetic, 0.182 real), which we attribute to typographic diversity in real documents — ALL CAPS names, diacritics, and multi-word surnames are underrepresented in the Faker-generated training distribution. Expanding training coverage of name formats is identified as the highest-priority data augmentation for future work.

V. EVALUATION

We evaluate Omni-Shield along four dimensions: PII detection accuracy against three baseline systems, agent-level contribution via ablation, error characterisation via confusion matrix analysis, and generalisation to real specimen documents. All experiments run on the same hardware used during training: an NVIDIA RTX 5060 Laptop GPU (8 GB VRAM), Intel Core i7, 16 GB RAM, Ubuntu 24.04.

A. Evaluation Protocol

a) Test set.: We evaluate on 150 synthetic identity card images (25 per country, seed 99999, held out from training). Each image is annotated with ground truth bounding boxes and PII values recorded at draw time. Five PII fields are annotated per card: Name, Date of Birth, ID Number, Face, and Signature. The test set is generated with the same pipeline as the training set to ensure visual consistency; statistical independence is guaranteed by the distinct random seed.

b) Metrics.: We compute precision, recall, and F1 for each of ten PII categories independently. A predicted bounding box is matched to a ground truth box using intersection-over-union (IoU) with a threshold of 0.3 for text fields and 0.1 for signature regions (which vary widely in spatial extent across card designs). For signature boxes, we additionally apply a containment check: if the ground truth box occupies at least 50% of the predicted box area, the prediction is counted as a true positive regardless of IoU. We report Macro F1 (mean over all categories) and Micro F1 (computed from aggregate TP, FP, FN counts). For text-only baselines and real document evaluation (where bounding boxes are unavailable), we use text value matching: a prediction is counted as a true positive if the predicted string is a substring of the ground truth value or vice versa, with Levenshtein distance tolerance ≤ 2 for strings of length ≥ 4 .

c) Evaluation framework.: All evaluations call the live FastAPI backend via HTTP (POST /redact/image, POST /redact/pdf). The evaluation scripts are fully standalone and are released alongside the system code.

B. Baseline Systems

We compare against three baselines representing the spectrum of available approaches.

Microsoft Presidio [1] is a mature open-source NLP-based PII detector. We apply EasyOCR to extract text from each test image and pass the extracted text to Presidio’s AnalyzerEngine with the `en_core_web_lg` spaCy model. Presidio operates on text only and cannot access pixel-level information.

Llama 3 8B (via Ollama) represents the approach of prompting a locally-hosted LLM to identify PII from OCR-extracted text. We use the same EasyOCR output and submit it with a structured prompt requesting a JSON array of detected PII entities. This baseline tests whether a larger general-purpose language model — $4\times$ the parameter count of our student VLM — can substitute for the specialised multi-agent pipeline.

Omni-Shield uses the full nine-agent pipeline with all PII categories enabled and no custom terms. The same 150 test images are processed in sequence with warm models (after first-request model loading).

C. Main Results

Table IV presents precision, recall, and F1 per PII category for all three systems.

The results reveal complementary strengths across systems. Llama 3 8B achieves the highest Macro F1 (0.270) and excels on text-based PII: Date of Birth (F1=0.854), ID Numbers (0.778), and Names (0.720). These strong results reflect Llama 3’s large parameter count and instruction-following capability, which enable reliable extraction of structured values from OCR text even when field labels vary across card designs.

However, Llama 3 scores F1=0.000 on Signatures and Faces — categories that require visual layout understanding rather than text analysis. Neither baseline can detect visual PII because both operate on OCR-extracted text rather than the document image.

Omni-Shield achieves the highest F1 on Signatures (0.870) and is the only system to detect this category at all. Signature detection is the single most important category by volume: ablation shows it accounts for a disproportionate share of total F1 (Section V-D). Omni-Shield also achieves competitive performance on Date of Birth (F1=0.522) — substantially below Llama 3 (0.854) but dramatically above Presidio (0.000) — reflecting the VLM’s ability to interpret layout context that pure NER misses.

Presidio achieves Macro F1=0.057, the lowest of the three systems. It attains non-zero F1 only on Names (0.290) and ID Numbers (0.280), where spaCy’s PERSON and structured pattern matchers fire on well-formed text tokens. It fails entirely on Signatures, Faces, Dates of Birth, and Addresses — five of the most prevalent PII categories on identity documents.

a) *The case for multi-modal redaction.*: The baseline comparison validates the central design claim of Omni-Shield: *text-only systems are insufficient for identity document redaction*. A deployment relying solely on Llama 3 — the strongest text baseline — would fail to redact every signature and face

photograph in the test set, leaving the most visually distinctive PII untouched. Conversely, a deployment relying solely on Omni-Shield would miss the text PII that Llama 3 handles well. The ideal system combines both approaches; we identify this hybrid as the primary direction for future work (Section VII).

D. Agent Ablation Study

To quantify each agent class’s contribution to overall performance, we run five pipeline configurations on the 150-card test set with different subsets of PII categories enabled.

Table V reports Macro F1 and Micro F1 for each configuration.

The full pipeline (Macro F1=0.158 on this 30-card subset; 0.187 on the full 150) establishes the baseline. Disabling Faces and Signatures reduces Macro F1 by 54.2% — the largest single drop of any configuration — confirming that visual PII detection drives the majority of the system’s value on identity documents. Disabling only Names reduces Macro F1 by 61.0%, reflecting that names are present on every card and that the pipeline’s name detection, while imperfect, contributes consistently across all 150 documents. Disabling Date of Birth and ID Numbers reduces Macro F1 by only 3.6%, consistent with the low recall observed for these categories in the full evaluation. Running only the text PII agents (Phones, Emails, Addresses) produces Macro F1=0.039 — barely above zero — because these fields are absent from almost all synthetic identity cards.

These results validate the multi-agent decomposition: each agent class contributes measurably, and no single component dominates to the degree that the others could be removed without significant loss.

E. Error Analysis

Figure 4 shows the PII category confusion matrix across 150 test documents. Of 498 total predictions, 262 are true positives, 175 are false positives, and 655 ground truth items are false negatives (missed detections).

False negatives dominate. The ratio of missed detections to false positives is approximately 4:1. The system is not confusing PII types with each other — when it detects a field, it classifies it correctly with high precision. The primary failure mode is recall: the VLM fails to generate predictions for fields it observes, particularly on cards where field positions deviate from the training distribution.

Signatures generate the most false positives relative to their true positive count (5 FP against 126 TP). Examination of failure cases reveals that the model occasionally identifies underline decorations or stamp regions as signature areas. The low FP rate relative to TP (3.8%) indicates the signature detector is well-calibrated.

Date of birth over-fires with 61 false positives against 65 true positives — a precision ratio of 51.7%. The confusion matrix shows that expiry dates, issue dates, and reference numbers containing date-like patterns account for the majority of DOB false positives. Adding explicit exclusion rules for

TABLE IV
PII DETECTION — COMPARISON WITH BASELINES

Category	Omni-Shield			Presidio			Llama 3 8B		
	P	R	F1	P	R	F1	P	R	F1
Names	0.60	0.21	0.31	0.35	0.24	0.29	0.96	0.58	0.72
Faces	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Signatures	0.85	0.90	0.87	0.00	0.00	0.00	0.00	0.00	0.00
Phone Numbers	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Email Addresses	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Date of Birth	0.51	0.54	0.52	0.00	0.00	0.00	0.79	0.94	0.85
ID Numbers	0.38	0.05	0.10	0.29	0.27	0.28	0.78	0.77	0.78
Addresses	0.11	0.05	0.07	0.00	0.00	0.00	0.42	0.30	0.35
Org Names	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Dates (General)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Macro F1	—	—	0.1867	—	—	0.0566	—	—	0.2703

TABLE V
AGENT ABLATION STUDY — MACRO F1

Configuration	Macro F1	Micro F1
All agents (baseline)	0.1584	0.3197
Faces + signatures disabled	0.0726	0.1493
Names only	0.0617	0.1289
DOB + ID numbers disabled	0.1527	0.3136
Phones + emails + addresses only	0.0392	0.0607

common non-PII date labels (*Valid Until*, *Issue Date*, *Expiry*) to the ContextAgent is the targeted fix.

Addresses exhibit a structural IoU problem. The model detects single address lines with correct precision, but ground truth annotations span the full multi-line address block. IoU between a one-line prediction and a three-line ground truth box is structurally low even when the detected region is spatially correct. Adopting a containment-based matching criterion (predicted box covers $\geq 50\%$ of ground truth area) for this category would substantially improve measured F1 without changing the underlying detection quality.

F. Per-Agent Latency

Table VI reports mean end-to-end latency per document format. For the image pipeline, Figure 3 breaks down latency by agent.

The LayoutAgent (VLM inference, 4.79 s mean) accounts for 46% of total pipeline time. OCRAgent (EasyOCR, 2.73 s) contributes 26%, and CriticAgent (LLM review, 1.85 s) 18%. All other agents together account for under 10% of total latency.

Two optimisation directions are immediately apparent. First, replacing EasyOCR with a lighter GPU-accelerated OCR engine (PaddleOCR, TrOCR) could halve OCR time. Second, replacing the CriticAgent’s full LLM inference pass with a lightweight heuristic classifier could eliminate its 1.85 s contribution with minimal accuracy impact, given that the critic’s primary function — removing organisational names and legal boilerplate — is well-suited to rule-based matching.

PDF, audio, text, Word, and PowerPoint documents bypass the VLM entirely, yielding latencies of 40 ms, 1.22 s, 10 ms, and sub-second respectively. This $225\times$ spread between the slowest

(image) and fastest (PDF/text) formats reflects the fundamental cost of visual understanding relative to text extraction.

G. Per-Country Analysis

Performance varies across the six countries in the test set. Germany achieves the highest Macro F1 (0.214), followed by India (0.196), Australia (0.164), USA (0.151), UK (0.126), and Canada (0.109). Canada also accounts for three of the four pipeline errors (HTTP 500 responses) in the full evaluation run, which we attribute to VRAM exhaustion on complex card layouts triggering an out-of-memory condition in the VLM inference kernel. These errors represent 2.0% of the test set and are resolved by restarting the model between consecutive complex requests.

The performance ordering does not correlate with training set size (equal at 333–334 per country) and instead appears to reflect visual distinctiveness: German and Indian cards use high-contrast field labels and consistent layouts that the VLM finds easier to parse, while UK and Canadian cards exhibit more varied typography and field positioning.

H. Real Specimen Evaluation

To assess generalisation beyond the synthetic distribution, we evaluate Omni-Shield on 19 real specimen identity documents sourced from Wikimedia Commons — officially published educational specimens containing no real PII. Documents were annotated by visual inspection; since pixel-level bounding boxes are impractical to obtain for real documents, we use text-value matching exclusively (IoU disabled, bbox = null).

Table VII compares per-category performance on synthetic versus real documents. Three findings stand out.

TABLE VI
MEAN REDACTION LATENCY BY DOCUMENT FORMAT

Format	Pipeline	Mean	Min	Max	N
Image (JPEG/PNG)	9-agent VLM pipeline	9.0s	6.3s	10.98s	5
PDF	Hybrid NER + PyMuPDF	0.04s	0.03s	0.05s	5
Audio (MP3/WAV)	Whisper STT + NER	1.22s	1.05s	1.78s	5
Plain text	Hybrid NER only	0.01s	0.01s	0.01s	5

TABLE VII
SYNTHETIC VS REAL SPECIMEN DOCUMENT EVALUATION

Category	Synthetic F1	Real P	Real R	Real F1
Faces	0.000*	0.826	1.000	0.905
Signatures	0.870	0.565	0.867	0.684
DOB	0.522	0.875	0.438	0.583
ID Numbers	0.096	1.000	0.200	0.333
Names	0.312	0.500	0.111	0.182
Addresses	0.067	0.000	0.000	0.000
Macro F1	0.187	–	–	0.269

*Synthetic faces F1=0 is an evaluation limitation (illustrated avatars), not a model failure.

Real document face recall = 1.0 confirms correct detection.

Macro F1 is higher on real documents (0.269 vs 0.187). This is the opposite of the expected synthetic-to-real degradation. We attribute this to the real specimen cards having more regular and visually prominent field labels than our synthetic test set, which introduced deliberate layout variation to stress-test the system.

Face detection transfers with near-perfect recall (F1 = 0.905, recall = 1.0). This confirms the YOLOv8n model correctly detects real photographic faces. The F1 = 0.000 reported in the synthetic evaluation is an evaluation artefact: synthetic cards use 2D illustrated avatars that COCO-trained YOLOv8 cannot detect as faces. A model file-name bug (`yolov8x.pt` referenced instead of the present `yolov8n.pt`) was identified and corrected during this evaluation; the live system has always applied face redaction correctly to real photographic faces.

ID number precision is perfect on real documents (P = 1.000). Every ID number prediction on real cards is a true positive. Recall remains low (0.200) — the model misses 16 of 20 instances — consistent with the synthetic finding that recall, not precision, is the primary gap for this category.

The largest synthetic-to-real performance drop is in Names (F1: 0.312 → 0.182), which we attribute to typographic diversity (ALL CAPS names, diacritics, multi-word surnames) underrepresented in the Faker-generated training distribution. Expanding the training corpus with more diverse name formats is identified as the highest-priority data augmentation for future work (Section VII).

VI. CRYPTOGRAPHIC AUDIT TRAIL

A redacted document is only trustworthy if a verifier can confirm *independently* that redaction occurred, when it occurred, and that the process was executed correctly. Manual audit logs are mutable and unverifiable; centralised databases require trust in their operator. Omni-Shield addresses this

through two complementary cryptographic mechanisms: a blockchain-anchored audit ledger that provides tamper-evident provenance, and a Groth16 zero-knowledge proof that attests to the correctness of the redaction without disclosing what was redacted. Both operate asynchronously — completing after the redacted document is returned to the user — and impose no latency penalty on the interactive redaction workflow.

A. Blockchain Audit Ledger

a) *Smart contract design.*: The `AuditLog.sol` Solidity contract maintains two on-chain data structures per record: a mapping (`bytes32 ⇒ Record`) for $O(1)$ hash lookup, and a per-owner mapping (`address ⇒ bytes32[]`) index that allows a wallet holder to enumerate all documents they have processed. Each `Record` stores the anchor hash, redaction count, block timestamp, and document type. The `addRecord()` function enforces an idempotency check — anchoring the same hash twice is silently skipped — and emits a `RecordAdded` event for off-chain indexing. The `verify()` view function is gas-free (read-only), enabling third-party verification without transaction cost.

b) *Salted hash commitment.*: A naive implementation would anchor `SHA256(document_bytes)` directly. This introduces a linkability vulnerability: identical documents processed multiple times produce identical on-chain hashes, allowing an observer monitoring the blockchain to detect repeated processing of the same document. Analysis of our audit log reveals 260 duplicate hashes among 424 records, all arising from test documents processed repeatedly during evaluation.

We address this by computing a salted anchor hash:

$$h_{\text{anchor}} = \text{SHA256}(h_{\text{doc}} \parallel \text{owner} \parallel \tau) \quad (1)$$

where $h_{\text{doc}} = \text{SHA256}(\text{document_bytes})$ is the original document hash, `owner` is the wallet address, and τ is the Unix

timestamp at anchor time. The original h_{doc} is preserved in the audit JSON for user-side verification (the user can recompute it from their document); h_{anchor} is what appears on-chain. The verification endpoint performs a two-step lookup: given h_{doc} from the user, it retrieves h_{anchor} from the stored audit record and queries the blockchain for that value.

Equation 1 ensures that even if two users process the same document, their on-chain anchors are distinct. An adversary who observes the blockchain learns only that *some* document was processed by *some* wallet at *some* time — no document identity information is recoverable from the on-chain record alone.

c) Deployment and gas cost.: The system currently deploys to a local Ganache Ethereum testnet [42] for development. Each `addRecord()` call consumes 264,333 gas units. At Ethereum mainnet prices (\$2,500 per ETH, 20 Gwei gas price), this corresponds to approximately \$13.22 per anchor — prohibitive for high-volume use. Deployment on Polygon L2 [30] reduces this to approximately \$0.006 per anchor at 30 Gwei and \$0.80 per MATIC, making production deployment economically viable. The `AuditLog.sol` contract is EVM-compatible and requires only an environment variable change to migrate between networks.

d) Verification performance.: The `verify()` view function resolves in 1.2 ms mean (measured over five queries to the local node), reflecting the $O(1)$ mapping lookup. This is fast enough to be embedded in document workflows without perceptible latency — a verifier can confirm a redaction record in under 2 ms, faster than a typical HTTP round-trip to a remote audit service.

B. Zero-Knowledge Proof of Redaction

a) Motivation.: The blockchain record proves that *some-one* anchored a hash at a specific time. It does not prove that the redaction was performed correctly — a malicious actor could anchor a hash corresponding to an unredacted document. A zero-knowledge proof closes this gap: it attests that the prover knows a set of redacted values consistent with the public commitment, without revealing the values themselves.

b) Circuit design.: We implement the ZK circuit in ZoKrates [33] using the Groth16 proving system over the BN128 elliptic curve. The circuit proves the following statement:

I know a private witness $\mathbf{w} = \{h_{\text{items}}\}$ such that $\text{Commit}(h_{\text{doc}}, h_{\text{items}}) = c_{\text{public}}$, where c_{public} is the publicly known commitment, h_{doc} is the public document hash, and h_{items} is the SHA-256 hash of the concatenated redacted values (known only to the prover).

The commitment function uses Pedersen hashing (available in the ZoKrates standard library) over the concatenation of the document hash bits and the items hash bits:

$$c = \text{PedersenHash}(h_{\text{doc}} \parallel h_{\text{items}}) \quad (2)$$

The circuit takes as *public inputs*: the document hash h_{doc} (as eight 32-bit unsigned integers), the redaction count, and the

commitment c (two 32-bit words). The circuit takes as *private input* (the witness): h_{items} (eight 32-bit unsigned integers). The private input never leaves the proving system and is not included in the proof.

c) Proof generation and verification.: Proof generation via ZoKrates `generate-proof` takes 20.3 s mean on the target hardware. This is executed asynchronously in a background thread immediately after the redacted document is saved; the user receives their output before proof generation completes. The proof becomes available in the audit record within approximately 25 s of the redaction request.

Proof verification uses ZoKrates `verify` against the on-disk verification key generated during the one-time trusted setup. Verification takes 22.1 ms mean — fast enough for interactive verification workflows.

Groth16 proofs are constant size (128 bytes for the proof object, three field elements for public inputs) regardless of the number of redacted items or the document size. This makes proof storage and transmission trivially cheap.

d) Trusted setup.: Groth16 requires a one-time trusted setup ceremony to generate the proving key (used by the prover) and verification key (used by verifiers). The current implementation uses a locally generated setup, which introduces a trust assumption: the party who ran the setup could theoretically generate false proofs for arbitrary statements. A production deployment should replace the local setup with a publicly verifiable multi-party computation (MPC) ceremony, or migrate to a transparent proof system (PLONK, STARKs) that eliminates the trusted setup requirement entirely. We identify this as a priority for future work (Section VII).

e) What the proof guarantees.: The ZK proof provides a *selective disclosure* property: an auditor can verify that a specific document was redacted and that the redaction committed to a specific (undisclosed) set of PII values, without learning what those values were. This is strictly stronger than the blockchain record alone, which only proves that a hash was anchored. Combined, the two mechanisms provide:

- **Tamper evidence:** the document hash is immutable on-chain; any alteration to the redacted output invalidates the record.
- **Provenance:** the owner wallet address and timestamp are permanently associated with the record.
- **Redaction correctness:** the ZK proof attests that the committed PII values are consistent with the claimed document.
- **Privacy:** neither the blockchain record nor the proof reveals which values were redacted or their content.

C. Audit Record Structure

Every completed redaction writes a `.audit.json` sidecar file alongside the output document. The sidecar contains:

- `sha256_hex`: original document hash (user-verifiable by re-hashing the input)
- `anchor_hash`: salted on-chain hash
- `anchor_salt`: the salt string used (required to recompute `anchor_hash` for verification)

- `redaction_count`: number of PII items redacted
- `coverage_pct`: percentage of document area or text volume redacted
- `pii_filter`: the category selection active at redaction time
- `redactions`: list of redacted items with masked values and bounding boxes
- `zk_proof`: Groth16 proof object, public inputs, and commitment
- `timestamp`: Unix timestamp of redaction
- `doc_type`: inferred document type

The sidecar is written synchronously before blockchain anchoring and proof generation begin, ensuring it is always available even if the asynchronous background tasks fail. A `GET /api/audit/{sha256}` endpoint serves sidecar data by hash prefix, enabling the frontend to retrieve and display the full audit record for any previously processed document.

D. Threat Model

We consider three classes of adversary.

Passive blockchain observer. An adversary who can read all on-chain transactions learns: which wallet addresses have processed documents, how many items were redacted, and when. They learn nothing about document content, PII values, or which documents were processed (because anchor hashes are salted). This is a substantial improvement over cloud-based alternatives, where the service provider learns the document content in full.

Malicious prover. An adversary who controls the proving system and wishes to generate a false proof (claiming a document was redacted when it was not) cannot do so without knowledge of the trusted setup trapdoor. Under the assumptions of the Groth16 proof system (computational soundness under the knowledge of exponent assumption), such forgeries are computationally infeasible. The local trusted setup weakens this guarantee: if the setup party is malicious, false proofs are possible. The MPC ceremony mitigation eliminates this assumption.

Compromised client. An adversary who compromises the user’s device after redaction can access the audit sidecar, including the anchor salt and redaction list. They cannot alter the on-chain record retroactively. The sidecar should be treated with the same access controls as the original document.

a) Out of scope.: We do not address: network-level attacks on the local Ganache node (protected by firewall in production), adversarial inputs designed to evade the VLM, or side-channel attacks on the GPU inference process. These are identified as areas for future security analysis.

E. Energy and Cost Analysis

Table VIII reports energy consumption and estimated carbon footprint for Omni-Shield and comparable cloud alternatives. At 100 W total system draw (RTX 5060 75 W plus 25 W overhead), Omni-Shield consumes 0.289 kWh per 1,000 image redactions — comparable to AWS Comprehend (estimated

0.200 kWh/1K) and substantially lower than GPT-4 Vision (estimated 1.000 kWh/1K).

Carbon footprint is highly sensitive to grid carbon intensity. On the Indian grid (0.82 kg CO₂/kWh), Omni-Shield produces 0.237 kg CO₂e per 1,000 redactions. On a renewable electricity tariff (0.08 kg/kWh), this falls to 0.023 kg — 1.3× lower than AWS Comprehend on its renewable energy commitment.

The additional energy cost of local-first processing relative to a cloud NLP API is 0.089 kWh per 1,000 documents, equivalent to approximately Rs.0.72 at Indian electricity retail rates. This is the quantified *privacy premium*: the energy overhead of the guarantee that sensitive identity documents never leave the device. Whether this premium is acceptable is a context-dependent policy decision; we report it as a measured quantity rather than an argument for or against local deployment.

VII. LIMITATIONS AND FUTURE WORK

We discuss the current limitations of Omni-Shield honestly, alongside concrete directions for future work.

A. Detection Limitations

a) Low recall on ID numbers (F1=0.096).: ID number detection exhibits the largest gap between precision and recall in our evaluation (P=0.381, R=0.055). The model correctly classifies an ID number when it generates a detection, but fails to generate detections on 94.5% of cards. We attribute this to two factors: (i) ID number fields use small fonts (8–10pt) that fall near the resolution threshold of the EasyOCR pipeline, producing fragmented OCR tokens that the VLM cannot confidently merge into a single field; and (ii) the six country-specific ID number formats (e.g., UK National Insurance, Indian Aadhaar, US Social Security) exhibit high visual variance, requiring more diverse training examples than the 2,000-card dataset provides. Future work will address this by expanding the training set with 500 additional cards targeting ID number field variation, and by adding a dedicated regex sweep post-processing step that applies country-specific patterns as a recall safety net.

b) Face detection not captured in evaluation.: Face bounding boxes are produced by a YOLOv8 object detection model operating independently of the VLM pipeline. Our evaluation framework measures PII detection via the `redactions` field of the API response, which does not include YOLOv8 face boxes. Consequently, the evaluation reports faces F1=0.000 despite the system correctly redacting face regions in the live application. A future evaluation pass will query face detections separately via a dedicated endpoint and evaluate IoU against ground truth face bounding boxes.

c) Address detection precision (P=0.113).: Address fields frequently span multiple text lines. Our ground truth annotations record the full multi-line bounding box, while the VLM typically generates a single-line detection. IoU between a one-line prediction and a three-line ground truth box is structurally low even when the detection is spatially correct. Future work will adopt a containment-based matching criterion for address fields (predicted box covers ≥50% of ground truth area) in place of strict IoU.

TABLE VIII
ENERGY CONSUMPTION AND CARBON FOOTPRINT PER 1,000 REDACTIONS

System	Type	kWh/1K	CO ₂ e (kg)	Grid assumed
Omni-Shield (image)	Local edge	0.289	0.237	India (0.82 kg/kWh)
Omni-Shield (image)	Local edge	0.289	0.113	US avg (0.39 kg/kWh)
Omni-Shield (image)	Local edge	0.289	0.023	Renewable (0.08 kg/kWh)
Omni-Shield (PDF)	Local edge	0.001	0.001	India
AWS Comprehend	Cloud NLP	0.200	0.030	AWS renewable
Google Cloud DLP	Cloud NLP	0.200	0.016	GCP renewable
GPT-4 Vision	Cloud LLM	1.000	0.150	AWS renewable

25W overhead). Cloud figures estimated from Patterson et al. 2022.

d) Date of birth over-firing (FP rate 48%).: The ContextAgent correctly identifies date-of-birth fields in most cases, but also flags expiry dates, issue dates, and reference numbers that contain date-like patterns. The confusion matrix confirms 61 false positives against 65 true positives for the DOB category. Adding explicit exclusion rules for common non-PII date labels (“Valid Until”, “Issue Date”, “Expiry”) to the ContextAgent heuristics is expected to reduce FP rate substantially.

B. Dataset and Generalization Limitations

a) Synthetic-only training and evaluation data.: Both the training set (2,000 cards) and the held-out test set (150 cards) are synthetically generated using Faker and Pillow. Synthetic data lacks the visual noise, printing artefacts, camera distortion, and handwritten annotations present in real identity documents. Transfer performance to real documents is unknown and likely lower than the F1 scores reported here. We identify testing on a curated set of 20–50 real anonymised identity documents as a high-priority future step; preliminary real-document experiments will be added before venue submission.

b) Six-country English-centric evaluation.: The current system supports six countries (Australia, Canada, Germany, India, UK, USA) with English-language field labels. German-language labels on German ID cards (e.g., *Geburtsdatum* for date of birth) are handled by the LABEL_TO_CATEGORY mapping but not systematically evaluated. Multilingual expansion using locale-specific Faker data and multilingual OCR is planned for a future version.

c) Per-country performance variance.: Macro F1 ranges from 0.109 (Canada) to 0.214 (Germany) across countries. Canada also accounts for three of the four pipeline errors (500 responses), attributable to VRAM exhaustion on complex card layouts. These errors represent 2.0% of the test set and are resolved by restarting models between requests; a future memory management improvement will implement proactive VRAM reclamation between pipeline runs.

C. Cryptographic Limitations

a) Hash linkability without salting.: SHA-256 is deterministic: identical documents anchored multiple times produce the same on-chain hash. An adversary monitoring the blockchain can infer that a specific document was processed repeatedly by observing repeated hash values. Analysis of our audit log reveals 260 duplicate hashes among 424 records, all

arising from re-processing the same test documents during evaluation. The mitigation is straightforward: compute the anchor hash as SHA-256(document_bytes || timestamp || owner_address) rather than SHA-256(document_bytes) alone. This countermeasure has been implemented: each anchor is salted with the owner address and a Unix timestamp, ensuring identical documents produce distinct on-chain hashes. The fix was validated empirically — two runs of the same document produced different anchor hashes while preserving the original SHA-256 for user verification.

b) Ethereum mainnet gas cost.: Each `addRecord()` call consumes 264,333 gas units, equivalent to approximately \$13.22 USD on Ethereum mainnet at \$2,500 per ETH. This is economically impractical for high-volume document processing. Deployment on Polygon L2 reduces the per-anchor cost to \$0.006 USD, which is viable for enterprise compliance workflows. The smart contract is EVM-compatible and requires only an environment variable change to migrate from Ganache to Polygon or Sepolia testnet.

c) ZK-SNARK proof generation latency.: Groth16/BN128 proof generation via ZoKrates requires 20.3 seconds (mean) on an RTX 5060 8GB VRAM, while on-chain verification completes in 22.1 milliseconds (mean), confirming that verification cost is negligible. Proof generation cost is absorbed by executing it asynchronously in a background thread; the user receives their redacted document before proof generation completes. Proof availability is therefore delayed relative to the redaction result. Future work will investigate STARKs or PLONK-based schemes that offer faster proving times at the cost of larger proof sizes, and will explore GPU-accelerated proving libraries such as `rapidsnark`.

d) Trusted setup requirement.: Groth16 proofs require a one-time trusted setup ceremony to generate the proving and verification keys. The current implementation uses a locally generated setup (no multi-party ceremony), which introduces a trust assumption: the party who ran the setup could theoretically generate false proofs. A production deployment should replace the local setup with a public multi-party computation (MPC) ceremony or migrate to a transparent proof system (STARK, Bulletproofs) that eliminates the trusted setup requirement entirely.

D. System and Deployment Limitations

a) *Single-document processing only.*: The current system processes one document at a time. Batch processing of document sets (e.g., a folder of scanned records) requires sequential API calls from the client. A native batch endpoint with job queuing, progress streaming, and a combined ZIP output is identified as the next major feature addition.

b) *Audio redaction replaces speech with silence.*: The audio pipeline silences detected PII segments rather than replacing them with a spoken “beep” or synthesized neutral tone. Silence can be as identifiable as the original speech in some contexts (e.g., a two-second gap where a name was spoken). Future work will add optional bleep-tone injection or text-to-speech replacement of redacted segments.

c) *Local deployment only.*: Omni-Shield is designed for local-first operation on consumer-grade hardware. No cloud deployment, containerised orchestration, or multi-user authentication is provided. Deployment to a remote server requires a Cloudflare tunnel or equivalent reverse proxy, and MetaMask wallet authentication currently uses a single deterministic Ganache account. A production-grade deployment would add OAuth or DID-based authentication and migrate blockchain anchoring to a public testnet.

E. Future Work Summary

We identify the following concrete directions for future work, ordered by expected impact:

- 1) **Hash salting for linkability resistance.** *Implemented.* Anchor hashes are now salted with owner address and Unix timestamp; validated empirically.
- 2) **Expanded training with ID number focus.** Generate 500 additional synthetic cards with deliberate ID number format diversity across all six countries and retrain the QLoRA adapter.
- 3) **Real document evaluation.** Collect 20–50 real anonymised identity documents and report transfer F1 alongside synthetic results.
- 4) **Separate face evaluation pass.** Add a YOLOv8 face evaluation module to the evaluation framework and report face F1 correctly.
- 5) **Polygon L2 deployment.** Replace Ganache with Sepolia testnet immediately; migrate to Polygon mainnet for production.
- 6) **GRPO reinforcement learning fine-tuning.** Apply Group Relative Policy Optimisation with an IoU-based reward signal from the SFT checkpoint to push bounding box precision beyond the current QLoRA adapter.
- 7) **Faster ZK proving.** Evaluate `rapidsnark` for GPU-accelerated Groth16 proving; alternatively migrate to PLONK for faster per-proof generation.
- 8) **Multi-party trusted setup.** Replace the local ZK setup with a public MPC ceremony or a transparent proof system to eliminate the trusted setup assumption.
- 9) **Batch processing endpoint.** Implement a native batch API with job queuing, progress streaming, and combined ZIP output.

- 10) **Multilingual expansion.** Add locale-aware field label mappings and multilingual OCR support for non-English documents.

VIII. CONCLUSION

We presented Omni-Shield, a local-first multi-modal PII redaction system that runs entirely on consumer-grade edge hardware and produces cryptographically verifiable audit records for every redaction decision. The system addresses a gap that existing tools leave open: no prior system simultaneously handles visual PII (signatures, faces), operates without transmitting documents to third-party infrastructure, and provides an independently verifiable audit trail with zero-knowledge proof of redaction correctness.

Summary of Contributions

Multi-modal nine-agent pipeline. We designed and implemented a nine-agent orchestration pipeline that processes six document formats on a single 8 GB GPU under 4-bit NF4 quantisation. The pipeline achieves 10.42 s mean latency for image redaction and under 50 ms for text and PDF documents. Ablation confirms that visual detection agents account for 54% of total F1 on identity documents, validating the multi-modal design over text-only alternatives.

Knowledge distillation for document PII. We fine-tune Qwen2-VL-2B-Instruct on 2,000 synthetically generated identity documents across six countries using QLoRA with teacher labels from the Claude API. The primary contribution of fine-tuning is output format consistency: the adapter trains the model to emit structured LABEL | VALUE | BBOX triples reliably, enabling downstream pipeline agents to parse and act on VLM output without heuristic recovery. Face detection generalises as an emergent capability post fine-tuning, despite faces not being explicitly labelled in the teacher annotations.

Cryptographic audit trail with ZK-SNARK. Every redaction is anchored on a local Ethereum blockchain using a salted SHA-256 commitment that prevents linkability attacks. A Groth16/BN128 proof generated by a ZoKrates circuit attests that the committed PII values are consistent with the document hash, without revealing what was redacted. Blockchain verification completes in 1.2 ms. ZK proof verification completes in 22.1 ms. Proof generation takes 20.3 s but is fully asynchronous, adding no user-facing latency. The system is the first, to our knowledge, to apply ZK-SNARKs to document redaction provenance.

Comprehensive evaluation on synthetic and real documents. We evaluate against Microsoft Presidio and Llama 3 8B on 150 annotated synthetic cards and 19 real specimen documents. Omni-Shield is the only system to detect signatures (F1=0.870) and faces (F1=0.905 on real documents), and achieves Macro F1=0.269 on real specimen cards — higher than the synthetic test result, indicating positive transfer from the synthetic training distribution.

Key Findings

Three findings from this work have broader implications for the design of privacy-preserving document systems.

Text-only systems are insufficient for identity document redaction. The strongest text-only baseline (Llama 3 8B, Macro F1=0.270) scores zero on signatures and faces — the two most legally significant visual PII categories. Any deployment that relies on text extraction alone will systematically fail to redact the most visually distinctive personal information on identity documents.

Local-first deployment is not a performance concession. Despite operating on a consumer GPU at 4-bit quantisation, Omni-Shield outperforms a mature cloud-backed NLP system (Presidio, Macro F1=0.057) by a factor of 3.3 on the same test set, and achieves competitive latency (10.42 s per image) against cloud vision APIs that require network round-trips for every document. The privacy premium of keeping documents on-device amounts to 0.089 kWh per 1,000 documents (approximately Rs.0.72 at Indian electricity rates) — a quantifiable and modest overhead.

Multi-modal agent pipelines are more robust than monolithic models. The CriticAgent pattern, ablation over agent configurations, and the TEXT-FIRST bounding box refinement strategy all demonstrate that decomposing the redaction task into specialised agents — each with a defined input, output, and failure mode — produces a system that is more debuggable, more ablatable, and more improvable than a single model would be. Each agent’s contribution is measurable; underperforming agents can be replaced without redesigning the pipeline.

Limitations

The primary limitations of the current system are: low recall on ID numbers (F1=0.096), a synthetic- only training distribution with documented transfer gaps for diverse name typography, a trusted setup requirement for the ZK circuit that weakens the proof’s soundness guarantee in the absence of a multi-party ceremony, and Ethereum mainnet gas costs that require L2 deployment for economically viable production use. These are discussed in detail in Section VII.

Future Work

The most impactful near-term improvements are: (i) replacing the ZK trusted setup with a public MPC ceremony or transparent proof system; (ii) expanding the training corpus with 500 additional cards targeting ID number format diversity to address the recall gap; (iii) combining Omni-Shield’s visual detection with a text-specialised model (such as Llama 3) in a hybrid pipeline that achieves high F1 across both visual and text PII categories; and (iv) deploying the blockchain contract on Polygon L2 for production-ready gas economics.

Reproducibility

The fine-tuned QLoRA adapter is available at [https://huggingface.co/\[anonymous\]/omni-shield-qwen2vl-pii-adapter](https://huggingface.co/[anonymous]/omni-shield-qwen2vl-pii-adapter). The evaluation harness, annotation pipeline, synthetic dataset generator, and all experiment scripts are available at [https://github.com/\[anonymous\]/omni-shield](https://github.com/[anonymous]/omni-shield). A Docker container with GPU support is provided for one-command

environment setup. All reported numbers are reproducible from the published code and data using the commands documented in the repository README.

Ethical Considerations

Omni-Shield is designed to protect privacy, not to circumvent it. The system does not retain documents after processing, does not transmit data off-device, and does not log PII values in plaintext beyond the masked audit record. The synthetic training data contains no real personal information. Specimen documents used in real-document evaluation are officially published educational materials containing no real PII. The blockchain audit trail is designed to provide accountability for operators of redaction systems, not to surveil the individuals whose documents are being processed. We release all artifacts under Apache 2.0 licence.

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